AI Messaging Sales Assistant

One AI assistant that answers customers on WhatsApp, Instagram and Messenger, books appointments, captures leads, and hands over to staff. Built for any business, with self-service sign-up.

Planning document, version 5 (concise)

1. Goal

Answer every customer message within seconds, 24 hours a day, in Chinese and English. Capture leads, book appointments, and hand over to a person whenever one is needed.

For	What it delivers
Customer	Instant replies in the chat app they already use, in their own language. A person is always one message away.
Staff	One inbox for all channels, an AI summary on every handover, and a simple way to give the chat back to the AI.
Business owner	A web admin panel to change prices, FAQ and AI behaviour without a developer, plus a daily summary on the phone.
Platform owner (you)	Businesses sign up and set up by themselves. You see usage and revenue, and can pause any business.

2. How it works

3. What the system includes

Module	What it does
Chat	One inbox for three channels. AI answers first, staff take over at any time. Assignment, labels, private notes, canned replies.
Customers	Searchable list with filters, tags, tiers and saved views. One customer ID across channels (matched by phone). Full chat and booking history.

Module	What it does
Bookings	Real availability from Google Calendar, **one calendar per practitioner or resource**. Book, reschedule, cancel, reminders.
Marketing	Template broadcasts and reactivation, only to customers who opted in.
Workflow	Automations (routing, reminders, follow-ups) edited visually in n8n.
Reports	Chats, leads by score, bookings, handovers, reply time, top questions.
Team	Staff accounts, roles, automatic assignment, routing by skill.
Settings	AI behaviour, catalog, resources, FAQ, hours, alerts, channel connections, plan and billing.

Smart extras

- **AI suggested replies** for staff, sent with one click.
- **Auto summary, tags, intent and mood** on every chat.
- **Translation view** so staff read and reply across Chinese and English.
- **Click-to-chat link and QR code** that also records where each customer came from.
- **After-hours mode**, **self-service rescheduling** in chat, and a **weekly owner insights** report.
- Later: aftercare and review requests, rebooking reminders, no-show recovery, waiting list, knowledge import from a price list or web page, review queue for unsure answers, deposit links.

4. For every industry, with self-service sign-up

Building block	How it fits any business
Universal base	Replies, FAQ answers, lead capture and scoring, handover. Works even with no booking.
Catalog	Services or products with name, price and description in each language.
Booking types	Appointment with a resource, table, viewing, class, quote visit, or none.
Custom lead questions	The owner chooses what to ask (budget, party size, location and so on).
Starter templates	Clinic, dental, salon, real estate, restaurant, education, repair, retail and more. Pre-fill everything, fully editable.
AI-assisted setup	Drafts the setup from a description, website or price list. The owner approves before going live.

Sign-up journey

1	Sign up and verify email
2	Describe the business, or paste the website or price list
3	AI drafts the setup, owner reviews and approves
4	Connect Google Calendars and WhatsApp with guided steps and test buttons
5	Test chat, choose a plan, go live

5. Parts and tools

Part	Tool	Role
Channels	Meta WhatsApp, Instagram, Messenger APIs	Where customers write. Each business uses its own number and Meta account.
Inbox	Chatwoot (open source)	Staff inbox. One account per business, created automatically.
Automation	n8n (open source)	Routes messages, sends replies, handover alerts, reminders, reports.

Part	Tool	Role
AI service	TypeScript API service in the aichat repo + Claude	Prompt, business knowledge and tools. Replies in the customer's language.
Admin panel	Next.js web app (React)	Business settings, customers, bookings, reports. Platform-owner console. Theme colour and logo per business.
Calendar	Google Calendar API	Availability and bookings, one calendar per practitioner.
Database	PostgreSQL	All data, separated per business with row-level rules.
Sign-up and billing	Public site + Stripe or local provider	Registration, plans, usage limits, invoices.

6. Technical stack

All of our own code is TypeScript, in one repository, for both the web app and the server. Chatwoot and n8n run as ready-made Docker images, so they need no custom programming, only configuration.

Languages and frameworks

Part	Technology	Notes
Language and tooling	TypeScript (strict mode), Node.js 22, pnpm monorepo, ESLint	One codebase with shared types between the web app and the API.
API, AI and workers	Fastify (API), Zod (validation), BullMQ (background jobs on Redis)	Webhooks from Chatwoot and Meta, AI tool calls, booking, usage limits, reminders, billing events.
AI model	Claude through the Anthropic TypeScript SDK	Default model Claude Haiku 5.5, set by one setting and changeable per business or plan. Tool use for calendar, booking, leads, handover.
Web app (admin, sign-up, platform console)	Next.js, React, Tailwind CSS, shadcn/ui components	Polished, interactive screens. Colours are theme variables, so each business can pick its own colour and logo, plus dark mode.
Inbox	Chatwoot (Ruby on Rails, Docker image)	Configured through its API and webhooks. One account per business, created through its platform interface.
Automation	n8n (Node.js, Docker image)	Workflows stored as JSON files in the repo and imported on start.
Calendar	Google Calendar API, googleapis library	Service account. Each business shares its calendars with it.
Billing	Stripe API (or a local provider)	Subscriptions, usage limits, invoices, webhooks.

Data, infrastructure and operations

Area	Technology	Notes
Database	PostgreSQL 16 with Drizzle ORM	SQL-first, works well with row-level security keyed by business ID. Separate databases for Chatwoot, n8n and our service on one server.
Cache and queues	Redis	Needed by Chatwoot. Also used for rate limits and background jobs.
Packaging	Docker Compose	Whole stack in one file, easy to move between servers.
Web server and HTTPS	Caddy	Automatic certificates. Public address needed for Meta webhooks, through a tunnel such as ngrok during development.
Backups	Scheduled pg_dump to cloud storage	Per-business export and deletion available.

Area	Technology	Notes
Monitoring	Uptime Kuma, health checks in the admin dashboard	Alerts to Telegram or email.
Email	An email-sending service over SMTP	Verification emails and notifications.

Security and quality

Area	Approach
Login	Email and password with argon2 hashing, secure sessions, optional two-step login (TOTP) for owners.
Secrets	Meta tokens and Google credentials encrypted at rest. Secrets kept in environment files, never in git.
Isolation	Business ID on every table plus row-level security. Automated tests try to read across businesses.
Audit	Sign-ins, support access, pauses and deletions are logged.
Testing	Vitest for logic (booking slots, handover rules, limits), Playwright for end-to-end screens, sample-message tests for workflows. GitHub Actions runs checks on every change.
Repository layout	aichat repo: apps/web, apps/api, shared packages, Docker Compose, database migrations, n8n workflow JSON, templates, docs.

Decided: TypeScript for all our own code. **Choices to confirm:** login library (Auth.js or Better Auth), Stripe or a local payment provider, and the email-sending service.

7. Handover, safety and separation

The AI hands over to staff when

- The customer asks for a person.
- Complaint, anger, refund or payment problem.
- Medical, legal or safety concern. The AI never gives such advice.
- The question is outside the business's knowledge, or the AI is unsure.

Safety and separation

- Answers only from the business's own knowledge. Never invents prices, offers or availability.
- Tells customers it is a virtual assistant. Customer text is treated as untrusted.
- Every record carries a business ID, and database rules block cross-business reads.
- Encrypted credentials, two-step login for owners, audit logs, data export and deletion.
- Sign-up protection: email verification, bot check, limits, and a kill switch per business.
- If the AI service fails, the customer gets a polite message and staff are alerted.

8. Plans, hosting and costs

Item	Approach
Plans	Free trial, then paid monthly plans with limits on AI replies, staff and resources. At the limit the AI pauses, the owner is told, and messages still reach the inbox.
Hosting	Free hosting for development, the demo and an invite-only beta. Public sign-up needs a stable paid server with backups.
AI cost	Pay per use and small (Claude Haiku 5.5 by default). Counted per business and passed on through plans.
Meta fees	Billed by Meta directly to each business.
Other	Domain, email sending, and payment provider fees.

9. Build order

Step	What gets built
1	Accounts and Meta approvals (start first, they can be slow). Docker stack with multi-business base.
2	Core bot on WhatsApp: bilingual answers, leads, handover.
3	Booking with Google Calendar, reminders.
4	Business admin panel, customers module.
5	Self-service sign-up, AI-assisted setup, plans and billing.
6	Platform-owner console, safety, legal pages. Instagram and Messenger.
7	Reports, hardening, smart extras, marketing and workflows.

Recommended path: run an invite-only beta first, then open public sign-up once Meta approvals and a paid server are in place.

10. Main risks

Risk	Reduction
Meta approvals delay launch	Start them first and run the beta meanwhile.
AI gives a wrong answer	Answer only from approved knowledge, handover rules, test chat, review queue.
Abuse or spam sign-ups	Verification, limits, caps, kill switch.
One business affecting another	Business ID everywhere, row-level rules, separation tests, own Meta account per business.
Legal and privacy duties	Terms, privacy policy, data export and deletion. Have a lawyer review for your country.

11. Decisions needed

- Platform name, domain, and the country billing starts in.
- Plan structure: trial length, limits, prices, payment provider.
- The company that owns the Meta business verification and signs the terms.
- Start with an invite-only beta, then open sign-up.
- Where it runs for now: free cloud VM (recommended) or an office machine.